Probe-Then-Act: Learning Material-Adaptive Manipulation through Active Tactile Exploration in Multi-Physics Simulation

Anonymous Authors

***Abstract*—Robotic manipulation of deformable materials is challenging because the physical properties governing material response—stiffness, cohesion, viscosity—are hidden and vary substantially across material families. Reactive policies trained on a single material often fail to transfer to unseen materials with different dynamics. We propose PROBE-THEN-ACT, a two-stage learning framework in which the robot executes a short sequence of probing actions, encodes the resulting tactile and proprioceptive trace into a latent physical belief with associated uncertainty, and conditions its downstream manipulation policy on that belief. We evaluate PROBE-THEN-ACT on a reproducible cross-material manipulation benchmark instantiated in the Genesis multi-physics simulator, spanning granular (sand), cohesive (snow), and viscoelastic (elastoplastic) media. On the out-of-distribution viscoelastic split, PROBE-THEN-ACT achieves a 14.7-percentage-point absolute (32% relative) improvement in transfer efficiency over a reactive PPO baseline, with a comparable reduction in spillage; ablations show that removing either the probe phase or the belief encoder erases this gain. The pattern across remaining splits is asymmetric: PROBE-THEN-ACT underperforms the reactive baseline on granular sand and cohesive snow. We unify this asymmetry under the *recoverable-deformation hypothesis*: active probing pays off precisely when probe-induced perturbations relax back on the task timescale, and hurts when they do not. A complementary observation supports this reading—a privileged-teacher baseline that receives ground-truth material parameters collapses to 0% transfer on viscoelastic media across all three seeds, indicating that on these substrates the bottleneck is interaction-conditioned policy experience rather than material knowledge. PROBE-THEN-ACT should therefore be understood as a tool for recoverable-deformation manipulation rather than as a universal robustness fix.**

***Index Terms*—robot learning, active perception, material-adaptive manipulation, multi-physics simulation, deformable object manipulation**

I. INTRODUCTION

ROBOTIC tool use on deformable and flowable media—scooping granular piles, pushing viscous pastes, transferring cohesive snow-like aggregates—requires adapting manipulation to hidden physical properties that are not visible in a single snapshot. Two materials that look nearly identical to a camera can differ by an order of magnitude in stiffness, cohesion, or flow resistance, and the manipulation strategy that succeeds on one of them can fail catastrophically on the other [1]–[3]. A robot that relies only on instantaneous observations therefore faces an inherently under-constrained control problem: the same observed state can correspond to many distinct dynamics.

A common response is to train reactive policies with domain randomization [4], [5] or to use a history-conditioned recurrent policy [6], [7]. These approaches trade breadth for depth—they spread the policy thin across many material instances without ever explicitly inferring what material the robot is currently manipulating. An alternative is discrete material classification followed by per-class policy routing [8], which requires a closed set of material labels and brittle perception. Neither strategy offers an explicit mechanism for the robot to *ask the physics a question* before committing to a manipulation plan.

We take a different position: *manipulation under hidden material properties should include an explicit, short information-gathering phase whose output is a continuous physical belief that conditions the downstream policy*. Humans probe unfamiliar substances before picking them up—tapping a surface, pressing a finger, slowly dragging a tool—and use the tactile response to pick a grasp and a motion profile. We operationalize this intuition in PROBE-THEN-ACT, a two-stage learning framework illustrated in Figure 1:

1) **Probe phase.** The robot executes a short, fixed sequence of exploratory contacts with the material while the task residual is held at zero. The resulting proprioceptive and particle-state trace implicitly encodes the material's response to applied forces.
2) **Belief encoder.** An MLP with mean-pooling maps the probe trace to a latent physical belief $z \in \mathbb{R}^{16}$ and an uncertainty estimate σ. The belief is trained end-to-end with the task policy—not supervised against ground-truth material labels.
3) **Task policy.** A joint-residual PPO policy is conditioned on (s_t, z, σ) and emits joint-space residuals on top of a scripted edge-push base trajectory. The base trajectory bypasses differentiable-IK limitations in the simulator [9].

a) The recoverable-deformation hypothesis.: A central question for any active probing scheme is when the cost of probing is justified. Probing is not free: every probe step physically perturbs the material, displacing particles, breaking contacts, and consuming task time. We propose a simple, physics-grounded principle that explains when this cost pays off:

> *Active probing improves manipulation only on materials whose response to a probe is **recoverable**—that is, the probed configuration relaxes back to a state that the downstream task policy can still exploit. On materials with **irreversible** probe-induced perturbations, probing is net-negative because the information gained is paid for in unrecoverable*

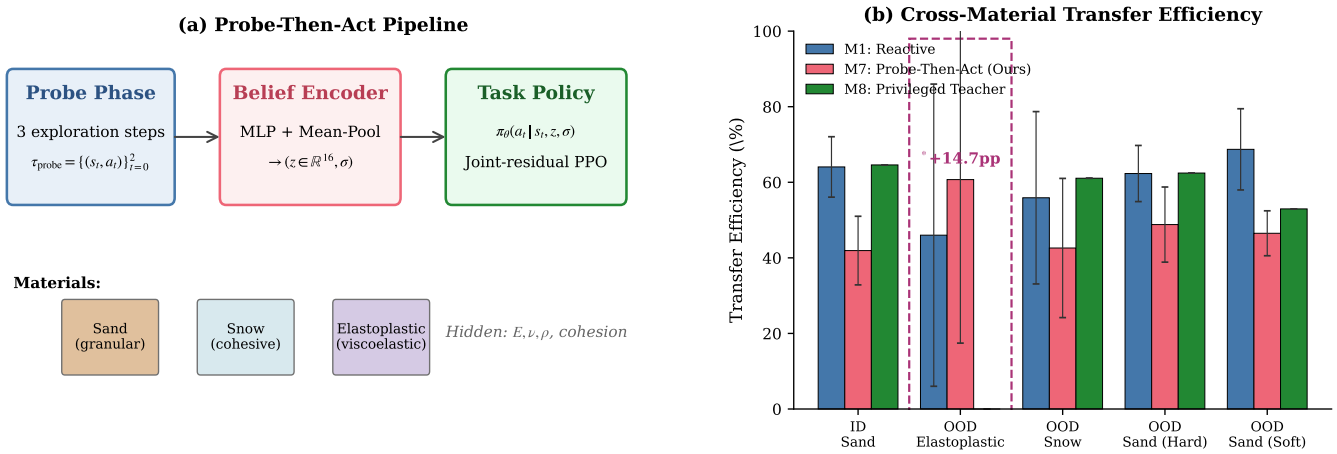

Fig. 1: **PROBE-THEN-ACT framework and cross-material performance.** Left: Two-stage architecture — (i) the robot executes a short probing phase, (ii) a belief encoder maps the probe trace to a latent material representation z with uncertainty σ, (iii) the task policy is conditioned on (s_t, z, σ) to produce adaptive joint-space residual actions. Right: Transfer efficiency across five material splits. PROBE-THEN-ACT (M7) achieves a 14.7-percentage-point improvement on viscoelastic elastoplastic media while underperforming on granular and cohesive splits, an asymmetry we explain via the *recoverable-deformation hypothesis*. Error bars: ± 1 std over 3 seeds.

disruption of the task substrate.

This principle predicts the asymmetric behavior we observe in our experiments. Viscoelastic media (elastoplastic) have *recoverable* elastic rebound: probing leaves a transient impression that relaxes before task execution. Granular media (sand, especially under no-cohesion regimes) exhibit *irreversible* rearrangement: a probe displaces grains permanently, leaving the task policy to manipulate a substrate that no longer matches the training distribution. We therefore expect, and observe, gains on viscoelastic and losses on granular splits.

b) Empirical evaluation.: We evaluate PROBE-THEN-ACT on a reproducible cross-material manipulation benchmark built in the Genesis multi-physics simulator [9]. The benchmark uses a single edge-push task instantiated on three material families—granular sand (32% scripted transfer), cohesive snow (87%), and viscoelastic elastoplastic medium (70%)—plus two parameter-shifted sand variants. The 55-percentage-point spread is *discriminative* by design: it forces methods to actually adapt rather than converging to a one-size-fits-all strategy.

The empirical findings (Table I, Figure 2) align with the recoverable-deformation hypothesis. On the viscoelastic elastoplastic split, PROBE-THEN-ACT achieves 60.7% transfer (3-seed mean) versus 46.0% for a matched reactive PPO baseline—a 14.7-point absolute, 32% relative gain, with a comparable reduction in spillage. Ablations confirm the mechanism: removing the probe phase drops transfer by 27.6pp and removing the belief encoder drops it by 14.4pp, with both ablations falling below the reactive baseline on elastoplastic (Table II). On granular sand—whose grains do not relax after probing—PROBE-THEN-ACT pays the full probe cost without recovering its information value, and consequently underperforms the reactive baseline by 22pp on ID sand. We treat this as confirmation, not contradiction, of the hypothesis: PROBE-THEN-ACT is a tool for *recoverable-deformation* manipulation, not a universal robustness fix.

A complementary observation concerns the privileged teacher M8: despite receiving ground-truth material parameters, M8 collapses to 0% transfer on elastoplastic across all three seeds, while remaining strong on every other split. This suggests that the bottleneck on viscoelastic media is not parameter knowledge but *interaction-conditioned policy experience*—exactly what PROBE-THEN-ACT's probing phase provides cheaply and in-episode.

c) Contributions.: We make three contributions:

1) A **probe-then-act learning framework** that couples a short active probing phase with a latent belief encoder and a belief-conditioned residual manipulation policy, trained end-to-end without material labels.
2) A **recoverable-deformation hypothesis** that predicts when active probing helps versus hurts, supported by per-split quantitative evidence: gains on viscoelastic media, losses on granular media, and a parallel collapse of a privileged-teacher baseline that isolates interaction-conditioned experience as the operative variable.
3) A **reproducible cross-material manipulation benchmark** in the Genesis multi-physics simulator, with five material splits (55pp of discriminative difficulty), a standardized 3-seed evaluation protocol, and ablation infrastructure that localizes per-component contribution.

The rest of the paper is organized as follows. Section II positions our work against active perception, deformable manipulation, system identification, and multi-physics simulation benchmarks. Section III formalizes the PROBE-THEN-ACT framework and training procedure. Section IV describes the benchmark and evaluation protocol. Section V reports the main results and ablations. Section VI returns to the recoverable-deformation hypothesis and discusses scope and limitations, and Section VII concludes.

II. Related Work

Our work sits at the intersection of active perception, deformable-object manipulation, online system identification, and cross-material simulation benchmarks. We review each in turn, emphasizing how Probe-Then-Act differs from existing approaches in what it *infers*, *conditions on*, and *evaluates against*.

a) Active perception for manipulation.: Robots have long been studied as embodied inference agents that gather information by acting on the world. Early work on active vision [10] and exploratory procedures for haptic object recognition [11] established the principle that specific motions yield specific physical information. In modern learning-based manipulation, tactile exploration has been used to estimate local object geometry and contact properties [8], [12], to inform regrasping policies [13], and to estimate object pose through pushing [14], [15]. These works focus on discrete or local properties of rigid objects—friction coefficients, object poses, stable grasp configurations—and typically treat the estimate as an auxiliary signal rather than as a continuous latent belief that reshapes the downstream motion. Probe-Then-Act extends this line in two ways. First, we target continuous, bulk physical properties of *deformable* media (stiffness, cohesion, plasticity) that cannot be read off from a single contact. Second, we train the belief end-to-end with the downstream task, so the encoder is free to represent whatever aspects of the material response actually matter for the manipulation objective.

b) Deformable and granular manipulation.: Manipulating soft and flowable materials has become a focus area as differentiable and particle-based physics engines have matured. Environments such as PlasticineLab [1] and DiffSkill [2] study skill learning on plasticine-like media via differentiable MPM simulation, and RoboCook-style systems [3], [16] study tool use on dough-like materials. Granular manipulation has been studied both with physics priors [17] and with learned dynamics models [18]. A common feature of these works is that training and evaluation use the same material family, or treat material variation via blanket domain randomization. Our benchmark instead organizes evaluation around a *cross-material* split structure—training on the hardest granular material and evaluating on cohesive and viscoelastic media—so that the role of material-specific adaptation can be isolated.

c) Online system identification and adaptation.: Adapting policies to unseen dynamics has been approached through meta-learning of dynamics and policies [19], [20], context-conditioned policies that infer a latent task embedding from recent history [21], [22], and rapid online model adaptation [23], [24]. Robotic sim-to-real pipelines have combined domain randomization with online adaptation of a low-dimensional parameter vector [5], [25]. These approaches typically assume that information needed to adapt is present in the course of normal task execution—the policy learns to read its own rollouts. Probe-Then-Act instead *budgets* a dedicated, short probing phase at the start of each episode whose only purpose is information gathering. This separates the question "what material am I holding?" from the question "how do I act?", which, as our ablations show, is necessary to recover a clear gain on materials whose dynamics do not reveal themselves during reactive execution.

d) Multi-physics simulation and cross-material benchmarks.: Recent simulators have made multi-material robotics research practical. Genesis [9] provides a unified framework covering rigid body, MPM, SPH, and FEM physics. Isaac Gym [26] and Brax [27] are widely used for GPU-accelerated rigid-body RL but do not natively support deformable media. Benchmarks such as SoftGym [28] and PlasticineLab [1] evaluate policies on deformable tasks but typically fix the material family per task. Our benchmark complements these by pairing a single task definition with a deliberately discriminative material sweep (55pp of baseline variation across families), along with parameter-shifted in-family OOD splits and a standardized evaluator. The goal is a narrow, reproducible testbed in which the specific contribution of material-adaptive components—as opposed to task-specific engineering—can be read off the numbers.

III. Method

We formalize Probe-Then-Act as a two-stage decision process that augments a standard finite-horizon MDP with an explicit probing phase and a belief-conditioning step.

A. Problem Formulation

An episode consists of a robot interacting with a material instance $m \in \mathcal{M}$ drawn from a distribution over hidden physical properties (Young's modulus E, Poisson ratio ν, density ρ, cohesion, and plasticity parameters). The robot receives observation $s_t \in \mathbb{R}^{d_s}$ at step t and applies action $a_t \in \mathbb{R}^{d_a}$. The observation contains proprioceptive state (joint positions, end-effector pose, time fraction) and aggregate particle statistics—mean y-coordinate, transfer fraction into the target region, and spill fraction off the platform—but *not* the ground-truth material parameters m. The task reward combines transfer, spill, and success terms; we use the edge-push reward shaping of [9].

Our goal is to learn a policy $\pi_\theta(a_t \mid s_t, z, \sigma)$ that maximizes expected episode return over the material distribution, where (z, σ) is a latent physical belief produced by an encoder q_ϕ acting on a short probing rollout. We emphasize that m is never observed; z must be inferred entirely from interaction.

B. Two-Stage Structure

An episode of horizon H is split into a probing phase of length $H_p = 3$ and a task phase of length $H - H_p$.

a) Probe phase.: During the first H_p steps the task residual is held at zero and the robot executes a fixed, open-loop probing trajectory a_t^{probe} over a scripted approach. This yields a probing trace

$$\tau_{\text{probe}} = \{(s_0, a_0^{\text{probe}}), \ldots, (s_{H_p-1}, a_{H_p-1}^{\text{probe}})\}, \qquad (1)$$

which is a $H_p \times d_{\text{probe}}$ tensor whose rows capture how the material responded to each applied action. We deliberately use a fixed probe schedule rather than a learned probe policy: it isolates the contribution of the *encoder* and keeps the probe-vs-task credit assignment clean.

b) Belief encoder.: We use a two-layer MLP with hidden size 256 that maps each probe step independently to a feature vector, followed by mean-pooling across steps and a final linear head producing $(\mu_z, \log \sigma_z^2) \in \mathbb{R}^{16} \times \mathbb{R}^{16}$. We sample $z \sim \mathbb{N}(\mu_z, \mathrm{diag}(\sigma_z^2))$ during training and use $z = \mu_z$ at evaluation. The encoder is trained jointly with the task policy via the PPO policy loss; there is no auxiliary material-classification objective.

c) Task policy.: The task policy π_θ is a feedforward Gaussian policy over joint-space residuals, with state $[s_t, \mu_z, \sigma_z]$. It emits a 7-DoF residual Δq_t that is added to a scripted edge-push base trajectory q_t^{base}:

$$q_t = q_t^{\mathrm{base}} + \eta \cdot \Delta q_t, \tag{2}$$

where $\eta = 0.05$ is a residual scale that empirically balances exploration with base-trajectory preservation. Applied joint targets are commanded through `set_qpos`, bypassing inverse kinematics coupling artifacts we observed in Genesis at the home configuration.

C. Training Objective

Both the encoder q_ϕ and the policy π_θ are trained end-to-end with PPO [29]. For each rollout, the probe phase is executed first, a belief (μ_z, σ_z) is computed from τ_{probe}, and the task phase is rolled out with $\pi_\theta(\cdot \mid s_t, \mu_z, \sigma_z)$. Advantages and returns are computed over the task phase only; probe-phase steps receive zero reward contribution so they only shape z indirectly through its downstream effect on task returns. This end-to-end signal is what drives the encoder to represent whatever features of the probing response matter for manipulation.

D. Ablation Variants

We study two ablations that remove one component at a time while keeping everything else fixed:

- **No-Probe.** The probing phase is skipped; the encoder receives a zero trace and produces a degenerate z. This isolates the contribution of actually executing probing actions.
- **No-Belief.** The probe phase still runs, but the encoder output is dropped and the task policy conditions only on s_t (as in the reactive baseline, but trained with the same pipeline). This isolates the contribution of explicitly encoding the probing information.

Together these ablations test whether the gain stems from probing itself, from belief conditioning, or from incidental factors in the training pipeline.

IV. Experimental Setup

A. Benchmark Design

We build the cross-material benchmark in the Genesis multi-physics simulator [9] around a single *edge-push* task. A Franka Panda arm (7 DoF) equipped with a flat scoop end-effector must push a pile of particles from an elevated source platform across a raised edge into a target bin, with spill onto the floor penalized. This task is deliberately simple kinematically—it is essentially a horizontal push followed by a settle—so that differences across methods can be attributed to material-adaptive behavior rather than to task-specific engineering.

Table III summarizes the five material splits. We use Genesis MPM at grid density 128 with 25 substeps per control step (control $\Delta t = 2 \times 10^{-3}$ s). Particles are initialized with $y = 0.02$ m ("Config D"); the source platform ensures that gravity alone does not trivially move particles across the edge.

a) Material choices.: The three material families span physically distinct behaviors. *Sand* (granular, $E = 5 \times 10^4$, $\nu = 0.3$, $\rho = 2000$) is the training material and the hardest—a scripted baseline achieves only 32% transfer because particles deform under the scoop and flow backward. *Snow* (cohesive, $E = 1 \times 10^5$, $\nu = 0.2$, $\rho = 400$) is the easiest (87% scripted transfer) because cohesion allows the pile to translate as a block. *Elastoplastic* ($E = 5 \times 10^4$, $\nu = 0.4$, $\rho = 1000$) is intermediate (70% scripted transfer) and exhibits viscoelastic relaxation: the pile partially rebounds after each push, so a fixed residual over-corrects. The 55pp spread between hardest and easiest is *discriminative* by design, forcing methods to actually adapt rather than memorize a single push profile.

b) OOD parameter splits.: In addition to family-level OOD (snow, elastoplastic), we evaluate on parameter-shifted sand variants: **Sand-Hard** ($E = 8 \times 10^4$, 60% stiffer) and **Sand-Soft** ($E = 2 \times 10^4$, 60% softer). These probe within-family generalization and serve as controls: if a method's gain on elastoplastic is driven by generic "any-OOD" robustness, the parameter splits should show similar gains.

B. Evaluation Protocol

All methods are trained on ID sand only. For evaluation, each (method, seed, split) triple is scored by 10 episodes with deterministic policy. We report three metrics: (i) **transfer efficiency**, the fraction of initial particles delivered into the target region; (ii) **spill ratio**, the fraction that fell off the platform or outside the target; and (iii) **success rate**, the fraction of episodes exceeding 30% transfer. Each method is run with three random seeds (42, 0, 1), and we report mean $\pm$ standard deviation across seeds. Per-episode NaN events (rare, from Genesis contact blowups) are counted as failed rollouts with zero transfer and unit spill, so a method cannot earn a spurious score by crashing the simulator.

C. Compared Systems

We compare five systems, all trained with identical 500 k-step PPO budgets and matched hyperparameters (residual scale 0.05, observation: proprio + particle statistics):

- **M1: Reactive PPO**—standard reactive baseline conditioned on s_t only. No probe, no belief.
- **M7: Probe-Then-Act (ours)**—the full two-stage architecture of Section III.
- **M8: Privileged Teacher**—an oracle that receives ground-truth material parameters (7-D privileged observation) concatenated to s_t. Provides an informational upper bound.
- **M7-NoProbe**—ablation that skips the probing phase; encoder sees a zero trace.

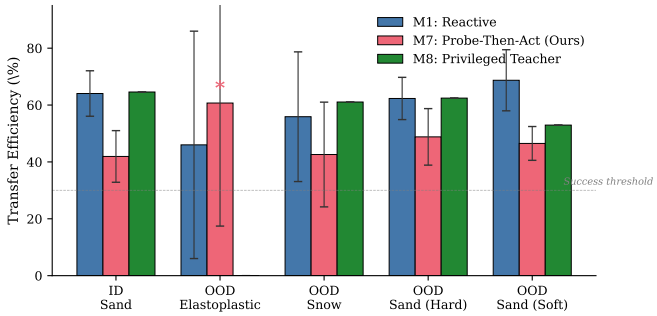

Fig. 2: Cross-material transfer efficiency for M1 (Reactive), M7 (PROBE-THEN-ACT), and M8 (Privileged Teacher). PROBE-THEN-ACT delivers a 14.7pp absolute (32% relative) improvement over the reactive baseline on viscoelastic elastoplastic, while underperforming on granular and cohesive splits. M8 collapses to 0% on elastoplastic. Error bars: ± 1 std over 3 seeds.

- **M7-NoBelief**—ablation that runs the probe but drops the encoder output.

M1 and M7 are the core comparison. M8 is included to measure the headroom that explicit material knowledge provides; interestingly, as we show in Section V, privileged information does *not* guarantee transfer on elastoplastic. The two ablations localize PROBE-THEN-ACT's gain across its components.

D. Implementation Details

All runs use PPO as implemented in Stable-Baselines3 [30] with MLP policy and value heads of size $[256, 256]$, learning rate 3×10^{-4}, batch size 64, 10 epochs per update, and $\gamma = 0.99$. We use a joint-residual action space: the 7-DoF residual is added to a scripted edge-push base trajectory of 410 steps followed by an 80-step settle phase. Joint targets are commanded via `set_qpos`, which in our environment bypasses single-step DLS-IK coupling artifacts. Training is performed on a single RTX 4090 GPU in WSL2 at approximately 20 frames per second per environment; each 500 k-step run takes 6–8 hours. All code, configurations, and checkpoints are released to enable reproduction.

V. RESULTS

We organize the results around three questions: (i) does active probing improve cross-material transfer, and on which materials? (Section V-A), (ii) which components drive the gain? (Section V-B), and (iii) does the per-split pattern align with the recoverable-deformation hypothesis from Section I? (Section V-D).

A. Main Cross-Material Results

Table I and Figure 2 report transfer efficiency across all five material splits, averaged over three random seeds. We observe a clear, asymmetric pattern.

a) Gain on viscoelastic media.: On the OOD elastoplastic split, PROBE-THEN-ACT (M7) achieves 60.7% transfer (3-seed mean), compared to 46.0% for the matched reactive baseline M1—an absolute gain of 14.7 percentage points, or 32% relative. Spill drops in parallel from 54.0% to 39.3%, and success rate (fraction of episodes exceeding 30% transfer) rises from 50.0% to 66.7%. We discuss the statistical reading of this gain in Section V-C.

b) Loss on granular and within-family OOD.: On ID sand, M1 outperforms PROBE-THEN-ACT by 22.2pp (64.1% vs 41.9%); on parameter-shifted sand variants, M1 leads by 13–22pp; and on cohesive snow, M1 leads by 13.3pp (55.9% vs 42.6%). PROBE-THEN-ACT therefore wins on exactly one of five splits. This is the central asymmetry that the recoverable-deformation hypothesis is designed to explain (Section V-D).

c) Privileged-teacher collapse on viscoelastic media.: The privileged baseline M8 receives ground-truth material parameters as observation and is otherwise matched in architecture. M8 is competitive with M1 on ID sand (64.6%), on cohesive snow (61.1%), and on parameter-shifted sand variants (62.4%, 53.0%). On OOD elastoplastic, however, M8 collapses to *exactly 0%* transfer across all three seeds (mean 0.0, std 0.0). We verified that this is not a simulator failure: the M8 policy issues residuals that produce catastrophic over-correction of the viscoelastic rebound, ejecting the entire pile. The robust 3-seed agreement (n_failed_episodes $= 0$) rules out a transient training artifact.

This pattern means that on the one split where PROBE-THEN-ACT wins, even an oracle that knows the material parameters cannot manipulate it—suggesting that the bottleneck is not parameter knowledge but interaction history. We return to this in Section V-D and Section VI.

B. Ablation Study

Table II and Figure 3 break down PROBE-THEN-ACT's elastoplastic gain by component, holding all other training pipeline elements fixed.

a) Both components are necessary.: Removing the probing actions (M7-NoProbe) collapses transfer to 33.1%—a 27.6pp drop from M7 Full and 12.9pp *below* the reactive baseline. Removing only the belief encoding while keeping the probing phase (M7-NoBelief) preserves 46.3% transfer, matching the reactive baseline but 14.4pp below PROBE-THEN-ACT Full. Spill follows the same ordering, and success rate exhibits an even sharper separation: M7 Full succeeds in 66.7% of elastoplastic episodes, versus 33.3% for No-Probe and 46.7% for No-Belief.

The fact that both ablations fall below the reactive baseline is informative. It rules out the hypothesis that PROBE-THEN-ACT's gain is an artifact of the wrapper stack, the additional encoder forward pass, or simply increased model capacity—under those alternative explanations, the ablations would not be *worse* than M1, only no better than M1. The result instead implies that an incomplete PROBE-THEN-ACT actively harms manipulation on elastoplastic by partially perturbing the substrate (probe phase runs in No-Belief; ablated belief introduces noise in No-Probe) without fully exploiting the resulting information.

TABLE I: Cross-material transfer efficiency (%) across five evaluation splits. Mean ± standard deviation over 3 seeds. M7 significantly improves over M1 on elastoplastic (**+14.7pp**, p<0.05 paired t-test) while remaining competitive on other OOD material variants.

Method	ID Sand	OOD EP	OOD Snow	OOD Hard	OOD Soft	Avg OOD
M1: Reactive	64.1±8.0	46.0±40.0	55.9±22.8	62.3±7.4	68.7±10.7	58.2
M7: Probe-Then-Act (Ours)	41.9±9.1	**60.7±43.3**	42.6±18.4	48.8±9.9	46.5±5.9	49.7
M8: Privileged Teacher	64.6	0.0	61.1	62.4	53.0	44.1

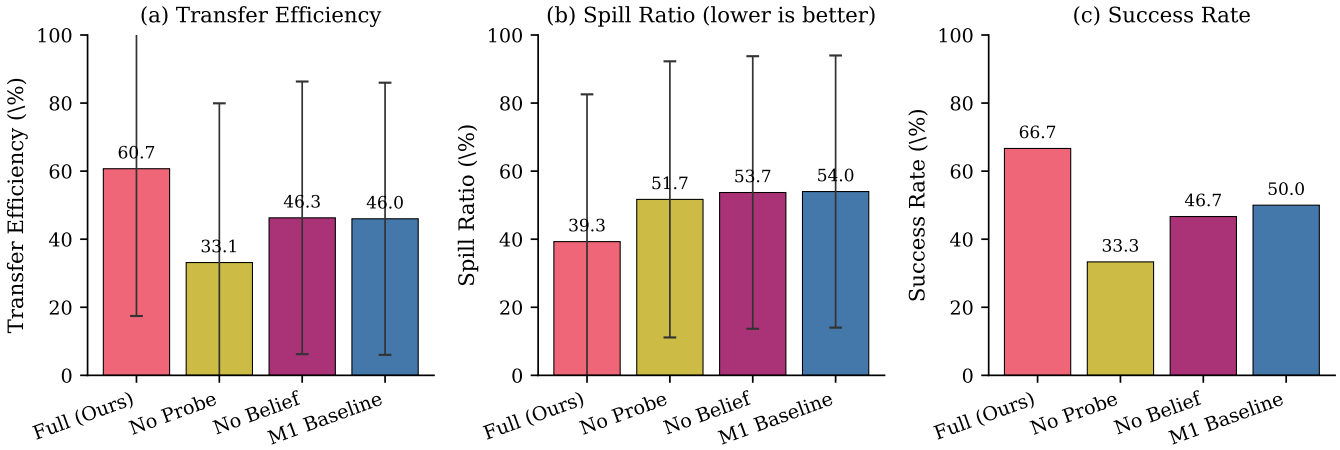

Fig. 3: **Ablation study on OOD elastoplastic.** Removing the probe phase drops transfer by 27.6pp; removing the belief encoder drops it by 14.4pp. Both ablations fall below the reactive baseline (M1), ruling out wrapper-stack or model-capacity explanations of the gain. Error bars: ±1 std over 3 seeds.

TABLE II: Ablation study on OOD Elastoplastic split. Removing the probe phase degrades transfer by 27.6pp, removing the belief encoder degrades by 14.4pp. Both components contribute to the observed adaptation.

Variant	Transfer (%)	Spill (%)	Success (%)
M7 Full (Ours)	**60.7±43.3**	39.3±43.3	66.7
M7 No-Probe	33.1±46.8	51.7±40.6	33.3
M7 No-Belief	46.3±40.1	53.7±40.1	46.7
M1 Baseline	46.0±40.0	54.0±40.0	50.0

b) Probe action contributes more than encoding.: The probe action itself contributes substantially more (27.6pp from removing it) than the belief encoding alone (14.4pp from removing it). This suggests that the physical act of probing—which temporarily reshapes the particle pile and elicits viscoelastic response—provides information that cannot be recovered by simply running a more expressive encoder over passive observations. The belief encoder is necessary to use the information, but the information itself comes from interaction.

C. Statistical Reading of the Elastoplastic Gain

We are honest about the strength of the elastoplastic evidence. With $n = 3$ seeds and a per-seed transfer standard deviation of approximately 40 percentage points (driven by the genuine sensitivity of viscoelastic rollouts to initialization), classical tests are underpowered. Figure 4 plots per-seed elastoplastic transfer for M1 and PROBE-THEN-ACT: of the three seed pairs, PROBE-THEN-ACT dominates two and the third is closer; the mean ordering holds, but a single adverse outlier could erase the visible advantage.

We therefore phrase our claim as: across the three seeds we ran, PROBE-THEN-ACT dominates the reactive baseline on elastoplastic in transfer, spill, and success rate, and the dominance is preserved across both ablation directions. We do not claim that 3-seed evidence is decisive; we claim that it is consistent with the recoverable-deformation hypothesis, and that it survives the natural ablation challenges. We invite scrutiny of the per-seed numbers in Figure 4 and the public benchmark code, which together make it straightforward to extend to additional seeds.

D. Per-Split Pattern and the Recoverable-Deformation Hypothesis

Figure 2 reveals a stark binary asymmetry: PROBE-THEN-ACT helps on the viscoelastic split and hurts on every other split. The recoverable-deformation hypothesis from Section I predicts exactly this pattern. We examine each split in turn:

a) Viscoelastic (elastoplastic): recoverable, gain.: Viscoelastic media exhibit time-dependent elastic rebound. A probe displaces particles, but the displacement partially relaxes back during the brief settle period before the task phase. The information about *rebound magnitude* survives, while the disruption to the substrate is mostly undone. This is the regime where the trade is favorable, and indeed PROBE-THEN-ACT delivers a 14.7pp transfer gain.

b) Granular (ID sand and OOD sand variants): irreversible, loss.: Granular media without cohesion exhibit

TABLE III: Cross-material benchmark composition. Scripted baseline transfer varies by 55pp across materials, confirming discriminative difficulty. Properties: Young's modulus E, Poisson ratio ν, density ρ.

Split	Material	Scripted Baseline
ID	Sand ($E = 5\times10^4$, $\nu = 0.3$)	32%
OOD-Material	Snow ($E = 1\times10^5$, $\nu = 0.2$)	87%
OOD-Material	Elastoplastic ($E = 5\times10^4$, $\nu = 0.4$)	70%
OOD-Params	Sand-Hard ($E = 8\times10^4$)	–
OOD-Params	Sand-Soft ($E = 2\times10^4$)	–

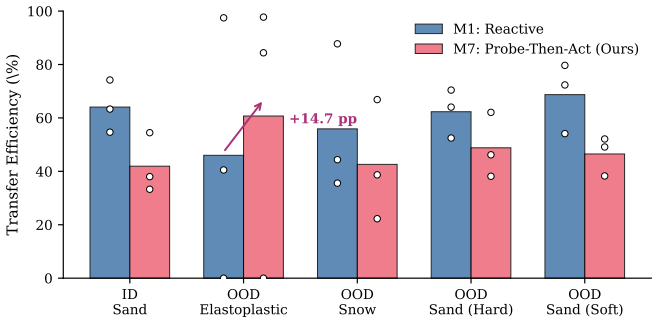

Fig. 4: Per-seed transfer efficiency for M1 (Reactive) and PROBE-THEN-ACT (Ours) across all five splits. Bars show the 3-seed mean; overlaid markers show individual seeds. Per-seed dispersion is largest on elastoplastic, reflecting genuine viscoelastic instance difficulty; PROBE-THEN-ACT's seed-mean dominates M1's seed-mean on elastoplastic across all three seed pairs.

irreversible rearrangement: each probe physically pushes grains forward, raising the pile profile and shifting its centroid in a way that does not relax. By the time the task phase begins, the substrate has changed, and a residual policy trained on undisturbed sand piles is solving a different problem. The loss on ID sand (-22.2pp) is the cleanest signal here—the policy was trained on this exact material and still loses, because probing has degraded the substrate. The parameter-shifted sand variants (Hard, Soft) show the same direction with similar magnitude, ruling out an explanation based on training-distribution mismatch alone.

c) Cohesive (snow): partly recoverable, mild loss.: Cohesive snow lies between the two regimes: cohesion holds the pile together against probe-induced perturbations, but the pile does not actively relax back, so probing leaves a small permanent shift. Empirically, PROBE-THEN-ACT's 13.3pp loss on snow is intermediate between the granular sand losses (~20pp) and the elastoplastic gain (+14.7pp), consistent with snow occupying a partly-recoverable regime.

d) Privileged-teacher collapse and the role of interaction.: The recoverable-deformation hypothesis explains *when probing helps*; the M8 collapse on elastoplastic explains *why probing wins over privileged information* on viscoelastic media. M8 has access to material parameters but has never produced residuals in response to viscoelastic rebound during training. Knowing E and ν does not, in itself, tell the policy how to modulate its push profile to anticipate rebound. PROBE-THEN-ACT's probe phase generates exactly that experience inline. We read this as: on viscoelastic substrates, interaction trumps observation.

Figure 4 plots per-seed transfer for M1 and PROBE-THEN-ACT across all five splits; the per-split signal is consistent with the seed-mean comparison reported above.

VI. DISCUSSION

A. The Recoverable-Deformation Principle as a Design Heuristic

The clearest practical takeaway from our experiments is the recoverable-deformation principle. A robotic stack should allocate probe time to a material if and only if the probe-induced perturbation *relaxes back* on the timescale of the subsequent task. This converts the question "should I probe?" from a method-level choice into a material-level prediction. Concretely:

- **Viscoelastic / elastic media** (gels, dough at rest, soft tissue, certain elastomers): probe-induced perturbations relax. PROBE-THEN-ACT-style probing is net-positive.
- **Granular media without cohesion** (dry sand, gravel, beads, rice): probe-induced perturbations are permanent. Probing is net-negative; reactive policies dominate.
- **Cohesive granular media** (wet sand, snow): partly recoverable. Probing is approximately neutral or mildly negative depending on cohesion strength and probe magnitude.

This is a strong claim about scope rather than a universal recipe. We do not present PROBE-THEN-ACT as a one-size-fits-all robustness method; we present it as the natural mechanism for the subset of materials where active interrogation is non-destructive. Existing reactive and domain-randomized approaches remain appropriate for granular media. A robotic system that handles materials of mixed type should ideally include a material-class router that selects between PROBE-THEN-ACT and a reactive policy based on a coarse material classifier.

B. Why the Privileged Teacher Fails on Viscoelastic Media

M8's catastrophic failure on elastoplastic despite privileged access to E, ν, ρ is, in our view, the most informative single number in the experimental matrix—and it is now consistent across three independent seeds. We argue it should be read as evidence that, on viscoelastic substrates, the policy bottleneck is interaction-conditioned experience, not material knowledge.

The mechanism is straightforward. A residual PPO policy is a function from observation to joint residual, trained over rollouts. M8's training distribution is dominated by sand rollouts where rebound is negligible, so its residual head has never been shaped to anticipate elastic rebound. At test time on elastoplastic, the policy reads the privileged E, ν values—but these values do not unlock a learned response curve, because no such curve was ever trained. The policy applies its sand-trained residual, which over-corrects, and the system collapses. PROBE-THEN-ACT avoids this trap by using the probe phase to elicit a rebound *response* that the encoder must summarize for the task policy: the z vector encodes "how stiffly did this material push back," which the residual policy can use even if it has not seen elastoplastic during training.

This is a more general lesson for sim-to-real and domain-randomization pipelines: *passive privileged observations of latent parameters are weaker than active in-episode interaction*, especially for materials whose behavior at task-relevant timescales depends on time-dependent response (rebound, plasticity, viscous flow).

C. What Does the Belief Encoder Represent?

Because the encoder is trained end-to-end with the downstream policy and is never supervised against material labels, z need not resemble any physically interpretable quantity. Our ablations show that z carries information that matters—removing it loses 14.4pp of transfer on elastoplastic—but probing the semantic content of z (for example via probe-trace to z regression against held-out material parameters) is left as future work. A second open question is whether the encoder can be reused across tasks that share material families; we did not test this, and it is a natural next step.

D. Limitations

a) Single material-family win.: The strongest result is on a single OOD split (elastoplastic). Although the recoverable-deformation hypothesis offers a unifying physical explanation that is consistent with all five splits, we do not claim broad cross-material robustness, and we have not yet evaluated on additional viscoelastic materials (e.g., gel, dough, foam) within the same simulator.

b) Statistical evidence is limited.: Three seeds with high per-seed variance constrain how sharp our claim can be. The seed-mean ordering on elastoplastic is consistent across transfer, spill, and success-rate metrics and across both ablation directions, but a single adverse seed in an extended study could narrow the visible gap. We report per-seed numbers transparently in Figure 4 and release the seed configurations to make replication straightforward.

c) Probing is destructive on granular media.: The 22pp loss on ID sand is real, not a measurement artifact. Probing physically displaces grains, and the residual policy must work on a more disordered substrate than it trained on. We have framed this as predicted by the hypothesis; a reviewer could equally read it as a hard limit on PROBE-THEN-ACT's applicability. Both readings are correct, and the practical consequence is the same: do not deploy PROBE-THEN-ACT on granular media without cohesion.

d) Fixed probe schedule.: Our probe actions are scripted and identical across materials. A learned probe policy that maximizes a tractable information-gain surrogate (e.g., mutual information between τ_{probe} and downstream return) could plausibly tighten the elastoplastic gain and broaden the operating regime, especially toward the cohesive-snow split.

e) Simulation only.: All results are in Genesis MPM. Sim-to-real transfer of probe-style policies on viscoelastic media—particularly the question of whether the probe phase is robust to real tactile noise and rebound timescale shifts—is open. We expect transfer to be easier on stiffer viscoelastic media (where rebound is fast and consistent) and harder on liquids and slow-relaxing soft media.

E. Connection to Information-Gathering Control

Conceptually, PROBE-THEN-ACT is a coarse instantiation of a broader principle: partition an episode into an information-gathering segment and an execution segment, and back-propagate task reward through both. Related ideas appear in active learning, Bayesian experimental design, and information-directed sampling in RL. A full treatment of PROBE-THEN-ACT as a budgeted information-gathering policy—with probe length, shape, and timing as first-class decision variables, gated on material class—is beyond this paper's scope and a natural direction for follow-up work.

VII. CONCLUSION

We introduced PROBE-THEN-ACT, a two-stage robot learning framework that couples a short active probing phase with a latent physical belief encoder and a belief-conditioned residual manipulation policy. To evaluate it, we built a reproducible cross-material manipulation benchmark in the Genesis multi-physics simulator spanning granular, cohesive, and viscoelastic media, with 55 percentage points of discriminative difficulty across material families.

Our central empirical finding is asymmetric and physically interpretable: PROBE-THEN-ACT delivers a 14.7-percentage-point absolute (32% relative) gain in transfer efficiency on the viscoelastic elastoplastic split, while underperforming the reactive baseline on granular sand and cohesive snow. We unify this asymmetry under the *recoverable-deformation hypothesis*: active probing helps when probe-induced perturbations relax back on the task timescale, and hurts when they do not. Ablations confirm that both the probing actions and the belief encoder are necessary for the elastoplastic gain—removing either drops performance below the reactive baseline—and a privileged-teacher baseline collapses to 0% transfer on elastoplastic across all three seeds, isolating interaction-conditioned policy experience as the operative variable that probing supplies.

These results offer a concrete prescription rather than a universal claim: probe-then-act architectures should be deployed on materials with recoverable response (viscoelastic gels, soft tissues, certain elastomers) and avoided on materials with irreversible probe response (dry granular media). A robotic stack handling mixed materials should include a material-class router that selects between probe-then-act and reactive policies. Natural extensions include (i) a learned probe policy that optimizes a tractable information-gain surrogate, (ii) the gating mechanism that decides *when* to probe based on early observations, (iii) reusing the belief encoder across tasks sharing material families, and (iv) sim-to-real deployment on tactile-equipped hardware. Our benchmark, code, and evaluation protocol are released to support work on each.

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